

CLIENT

React Frontend

- Analysis
- Results
- Report
- Workspaces
- Sessions
- Settings

REVERSE PROXY

nginx

ORCHESTRATOR · FASTAPI

Agent

Iterative LLM loop

Multi-turn tool calls

Session context

Task Runner

Background thread pool

Inference queue

Error explanation

Watcher Service

Filesystem polling

Triggers agent on new file

SSE console stream

Tool Registry

MCP client · discovers and calls tools at runtime

REST API + SSE

query · events · tasks · sessions · upload · workspaces

SQLITE

sessions

messages

tasks

files

workspaces

LLM BACKEND

Gemini API

Ollama

MCP SERVER

monai

stdio

Analyze image

List models

Download model

Run inference

MCP SERVER

radlex

stdio

List subspecialties

Find templates

Generate report

MCP SERVER

utils

stdio

Parse DICOM

Write file

List directory

Find files

MCP SERVER

cellpose

stdio

Segment cells 2D

Segment cells 3D

Save overlay

MCP SERVER

ipath

stdio

Fetch thumbnail

Get dimensions

Scale ROI

Fetch ROI

MCP SERVER

skills

stdio

Read skill

Execute script

Read references

MCP SERVER

fhir

HTTP

Patient records

FHIR standard

MONAI bundles

iPath platform